

Project II: Multiple Regression Modeling

King County House Sales

Your Name(s)

May 25, 2026

Summary of Data

```
house_df <- read_csv(
  "/Users/arezoog/Desktop/Spring 2026/STA135/Data/kc_house_data.csv"
)

summary_vars <- c(
  "price", "bedrooms", "bathrooms", "sqft_living",
  "sqft_lot", "floors", "waterfront", "view",
  "condition", "grade", "yr_built", "yr_renovated"
)

summary_df <- data.frame(
  Variable = summary_vars,
  Mean = sapply(house_df[, summary_vars], mean, na.rm = TRUE),
  SD = sapply(house_df[, summary_vars], sd, na.rm = TRUE),
  Minimum = sapply(house_df[, summary_vars], min, na.rm = TRUE),
  Maximum = sapply(house_df[, summary_vars], max, na.rm = TRUE)
)

summary_df[, -1] <- round(summary_df[, -1], 2)

kable(
  summary_df,
  caption = "Summary statistics for the key variables in the King County dataset",
  row.names = FALSE
)
```

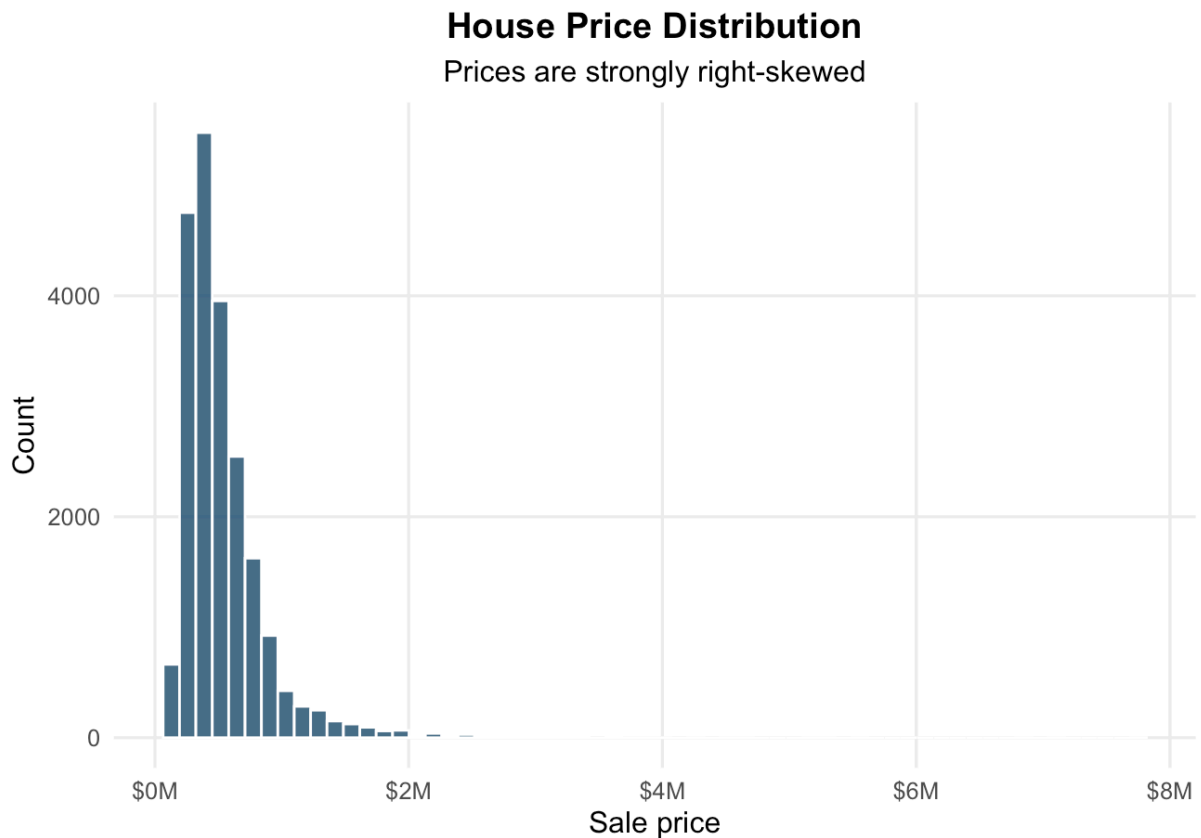
Summary statistics for the key variables in the King County dataset

Variable	Mean	SD	Minimum	Maximum
price	540296.57	367368.14	78000.0	7700000.0
bedrooms	3.37	0.93	1.0	33.0
bathrooms	2.12	0.77	0.5	8.0
sqft_living	2080.32	918.11	370.0	13540.0
sqft_lot	15099.41	41412.64	520.0	1651359.0

Variable	Mean	SD	Minimum	Maximum
floors	1.49	0.54	1.0	3.5
waterfront	0.01	0.09	0.0	1.0
view	0.23	0.77	0.0	4.0
condition	3.41	0.65	1.0	5.0
grade	7.66	1.17	3.0	13.0
yr_built	1971.00	29.38	1900.0	2015.0
yr_renovated	83.64	399.95	0.0	2015.0

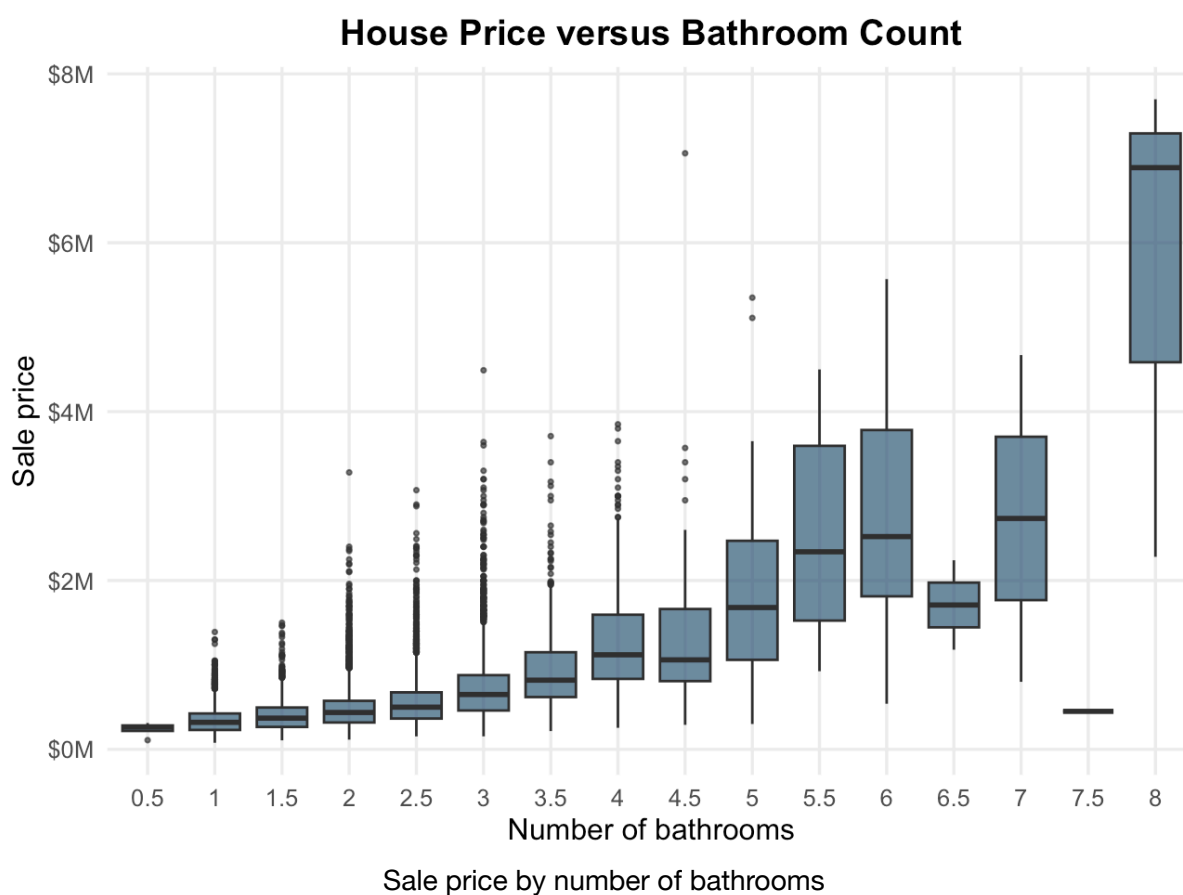
Explanatory Plot

```
ggplot(house_df, aes(x = price)) +
  geom_histogram(bins = 60, fill = "#1B4F72", color = "white", alpha = 0.85) +
  scale_x_continuous(
    labels = scales::label_dollar(scale = 1e-6, suffix = "M")
  ) +
  labs(
    title = "House Price Distribution",
    subtitle = "Prices are strongly right-skewed",
    x = "Sale price",
    y = "Count"
  )
```

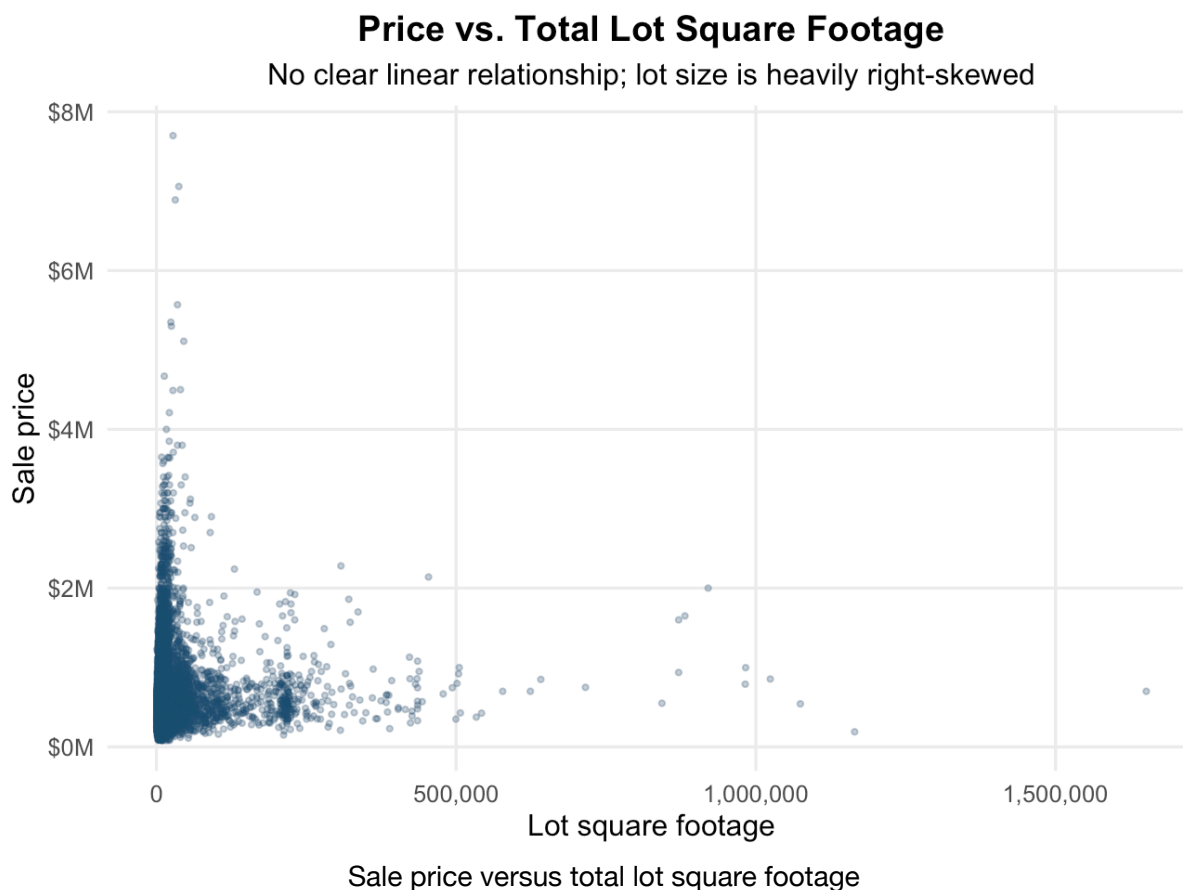


Distribution of home sale prices in King County (May 2014 – May 2015)

```
bath_df <- house_df %>%  
  mutate(bathrooms_f = factor(round(bathrooms * 2) / 2))  
  
ggplot(bath_df, aes(x = bathrooms_f, y = price)) +  
  geom_boxplot(fill = "#1B4F72", alpha = 0.7, outlier.size = 0.6) +  
  scale_y_continuous(  
    labels = scales::label_dollar(scale = 1e-6, suffix = "M")  
  ) +  
  labs(  
    title = "House Price versus Bathroom Count",  
    x = "Number of bathrooms",  
    y = "Sale price"  
  )
```



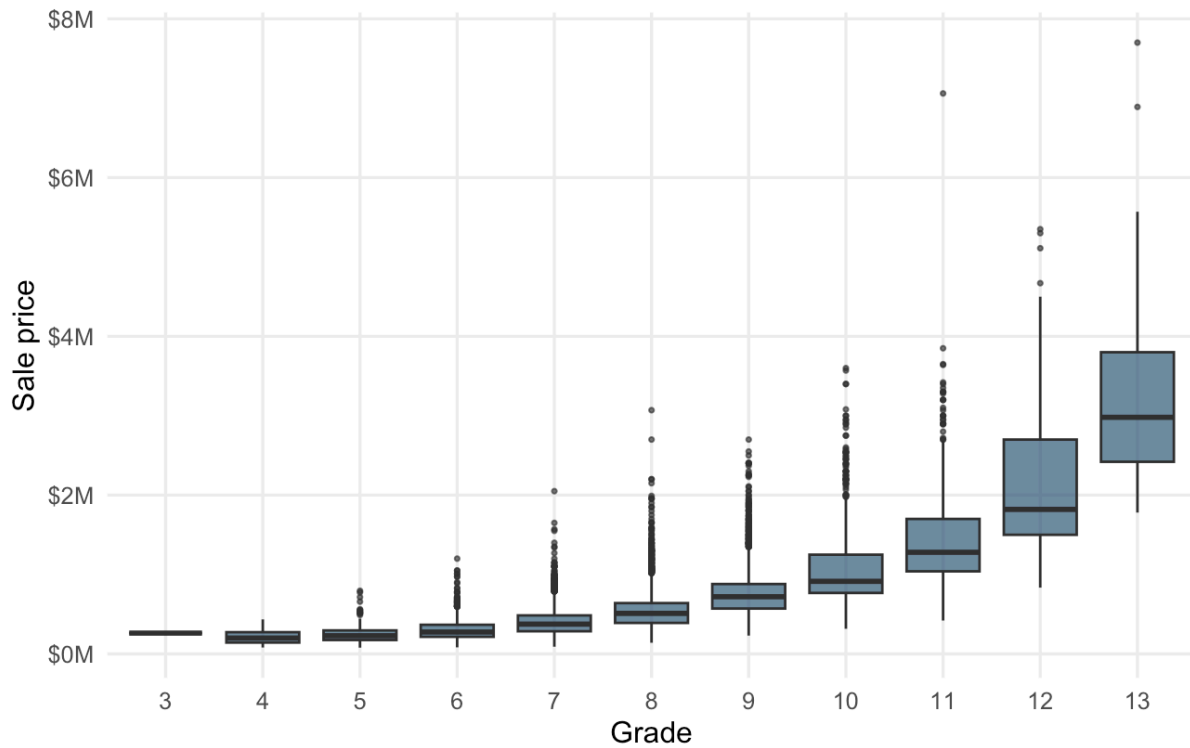
```
ggplot(house_df, aes(x = sqft_lot, y = price)) +
  geom_point(size = 0.8, alpha = 0.3, color = "#1B4F72") +
  scale_x_continuous(labels = scales::label_comma()) +
  scale_y_continuous(
    labels = scales::label_dollar(scale = 1e-6, suffix = "M")
  ) +
  labs(
    title = "Price vs. Total Lot Square Footage",
    subtitle = "No clear linear relationship; lot size is heavily right-skewed",
    x = "Lot square footage",
    y = "Sale price"
  )
)
```



```
ggplot(house_df, aes(x = factor(grade), y = price)) +
  geom_boxplot(fill = "#1B4F72", alpha = 0.7, outlier.size = 0.6) +
  scale_y_continuous(
    labels = scales::label_dollar(scale = 1e-6, suffix = "M")
  ) +
  labs(
    title = "House Price versus Construction Grade",
    subtitle = "Price increases with grade; relationship is non-linear",
    x = "Grade",
    y = "Sale price"
  )
)
```

House Price versus Construction Grade

Price increases with grade; relationship is non-linear



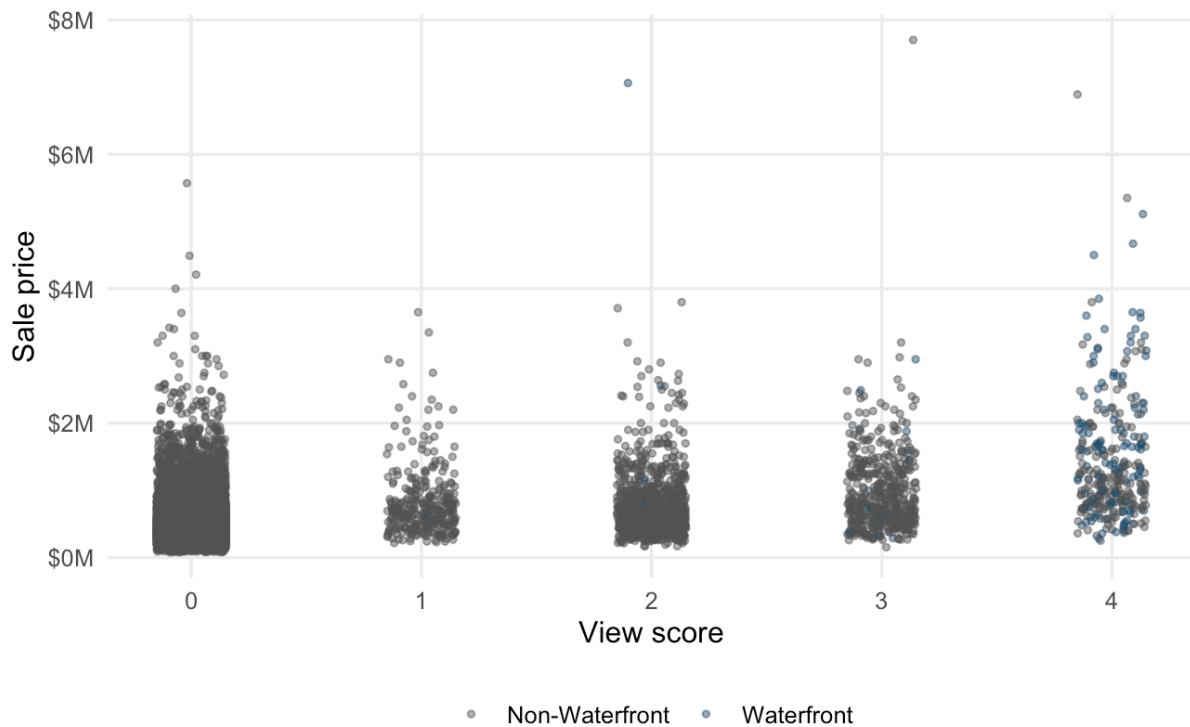
Sale price by construction grade

```
wf_df <- house_df %>%
  mutate(
    Waterfront = ifelse(
      !is.na(waterfront) & waterfront == 1,
      "Waterfront",
      "Non-Waterfront"
    )
  )

ggplot(wf_df, aes(x = view, y = price, color = Waterfront)) +
  geom_jitter(size = 1.0, alpha = 0.5, width = 0.15) +
  scale_color_manual(
    values = c("Non-Waterfront" = "#555555", "Waterfront" = "#1B4F72")
  ) +
  scale_y_continuous(
    labels = scales::label_dollar(scale = 1e-6, suffix = "M")
  ) +
  labs(
    title = "View Score vs. Price by Waterfront Status",
    subtitle = "Waterfront homes command higher prices at every view level",
    x = "View score",
    y = "Sale price",
    color = NULL
  )
```

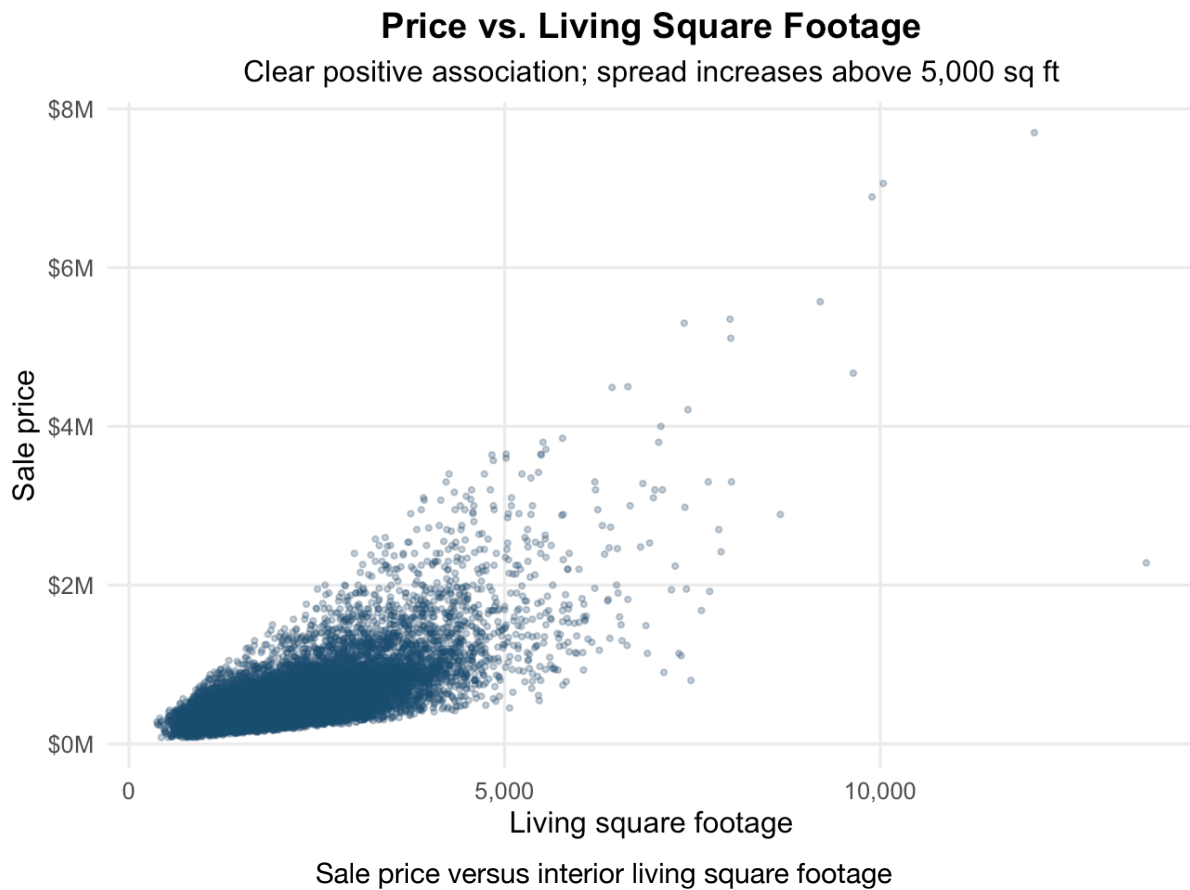
View Score vs. Price by Waterfront Status

Waterfront homes command higher prices at every view level



Sale price versus view score, colored by waterfront status

```
ggplot(house_df, aes(x = sqft_living, y = price)) +  
  geom_point(size = 0.8, alpha = 0.3, color = "#1B4F72") +  
  scale_x_continuous(labels = scales::label_comma()) +  
  scale_y_continuous(  
    labels = scales::label_dollar(scale = 1e-6, suffix = "M")  
  ) +  
  labs(  
    title = "Price vs. Living Square Footage",  
    subtitle = "Clear positive association; spread increases above 5,000 sq ft",  
    x = "Living square footage",  
    y = "Sale price"  
  )
```



Data Cleaning and Distance Feature

```
df <- as.data.frame(house_df)

df$sqft_basement <- suppressWarnings(as.numeric(as.character(df$sqft_basement)))
df$sqft_basement[is.na(df$sqft_basement)] <- 0

df$yr_renovated[is.na(df$yr_renovated)] <- 0
df$waterfront[is.na(df$waterfront)] <- 0

df <- df[!is.na(df$view), ]

house_coords <- cbind(df$long, df$lat)
downtown_coord <- c(-122.334862, 47.604942)

df$ft_to_downtown <- distHaversine(house_coords, downtown_coord) * 3.28084

drop_cols <- c("id", "date", "sqft_living15", "sqft_lot15", "zipcode", "lat", "long")
house_df_clean <- df[, !(colnames(df) %in% drop_cols)]
```

Model Fitting

```

# Full model
full_model <- lm(
  price ~ bedrooms + bathrooms + sqft_living + sqft_lot + floors +
    waterfront + view + condition + grade + sqft_above +
    sqft_basement + yr_built + yr_renovated + ft_to_downtown,
  data = house_df_clean
)

# Remove identified outliers by index (from diagnostics)
outlier_rows <- c(15468, 3833, 7082)
house_df_final <- house_df_clean[-outlier_rows, ]

# Final selected model (no sqft_above, sqft_basement, distance)
final_model <- lm(
  price ~ bedrooms + bathrooms + sqft_living + sqft_lot + floors +
    waterfront + view + condition + grade + yr_built + yr_renovated,
  data = house_df_final
)

```

```

model_summary <- summary(final_model)
final_tidy <- tidy(final_model, conf.int = TRUE)
final_glance <- glance(final_model)

r_sq <- final_glance$r.squared
adj_r <- final_glance$adj.r.squared
f_stat <- final_glance$statistic
f_df1 <- final_glance$df
f_df2 <- final_glance$df.residual

```

Analysis


```

coef_table <- final_tidy[, c(
  "term", "estimate", "std.error",
  "statistic", "p.value", "conf.low", "conf.high"
)]
names(coef_table) <- c(
  "Term", "Estimate", "Std. Error",
  "t statistic", "p-value", "95% CI lower", "95% CI upper"
)

coef_table[, c(
  "Estimate", "Std. Error", "t statistic",
  "95% CI lower", "95% CI upper"
)] <- round(
  coef_table[, c(
    "Estimate", "Std. Error", "t statistic",
    "95% CI lower", "95% CI upper"
  )],
  2
)

coef_table$p-value <- format_p(as.numeric(final_tidy$p.value))

kable(
  coef_table,
  caption = "Estimated coefficients for the final model",
  align = c("l", rep("r", 6))
)

```

Estimated coefficients for the final model

Term	Estimate	Std. Error	t statistic	p-value	95% CI lower	95% CI upper
(Intercept)	6174672.36	135748.05	45.49	<0.001	5908596.12	6440748.61
bedrooms	-38869.45	2034.77	-19.10	<0.001	-42857.75	-34881.15
bathrooms	45529.06	3445.89	13.21	<0.001	38774.86	52283.26
sqft_living	169.16	3.28	51.50	<0.001	162.72	175.60
sqft_lot	-0.26	0.04	-7.18	<0.001	-0.33	-0.19
floors	24409.28	3426.73	7.12	<0.001	17692.63	31125.93
waterfront	585370.05	19516.12	29.99	<0.001	547117.02	623623.09
view	47101.09	2207.11	21.34	<0.001	42774.98	51427.19
condition	19435.22	2487.25	7.81	<0.001	14560.02	24310.41
grade	125269.65	2143.82	58.43	<0.001	121067.60	129471.71
yr_built	-3564.45	69.76	-51.10	<0.001	-3701.18	-3427.72
yr_renovated	13.05	4.26	3.06	0.0022	4.70	21.39

```

model_living <- lm(
  price ~ bedrooms + bathrooms + sqft_living + floors +
    waterfront + view + condition + grade + yr_built + yr_renovated,
  data = house_df_final
)

model_lot <- lm(
  price ~ bedrooms + bathrooms + sqft_lot + floors +
    waterfront + view + condition + grade + yr_built + yr_renovated,
  data = house_df_final
)

adj_r2_living <- round(summary(model_living)$adj.r.squared, 4)
adj_r2_lot <- round(summary(model_lot)$adj.r.squared, 4)

t_living <- round(coef(model_summary)["sqft_living", "t value"], 2)
t_lot <- round(coef(model_summary)["sqft_lot", "t value"], 2)

```

```

model_no_grade <- lm(
  price ~ bedrooms + bathrooms + sqft_living + sqft_lot + floors +
    waterfront + view + condition + yr_built + yr_renovated,
  data = house_df_final
)

pf_grade <- anova(model_no_grade, final_model)
grade_f <- round(pf_grade[2, "F"], 2)
grade_df1 <- pf_grade[2, "Df"]
grade_df2 <- pf_grade[2, "Res.Df"]
grade_p_fmt <- format_p(pf_grade[2, "Pr(>F)"])

kable(
  pf_grade, digits = 4,
  caption = "Partial F-test: reduced model without grade versus full model"
)

```

Partial F-test: reduced model without grade versus full model

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
21520	1.162699e+15	NA	NA	NA	NA
21519	1.003478e+15	1	1.592207e+14	3414.395	0

```

model_interaction <- lm(
  price ~ bedrooms + bathrooms + sqft_living + sqft_lot + floors +
    waterfront * view + condition + grade + yr_built + yr_renovated,
  data = house_df_final
)

pf_interact <- anova(final_model, model_interaction)
int_f <- round(pf_interact[2, "F"], 2)
int_df1 <- pf_interact[2, "Df"]
int_df2 <- pf_interact[2, "Res.Df"]
int_p_fmt <- format_p(pf_interact[2, "Pr(>F)"])

interact_ci <- confint(model_interaction)["waterfront:view", ]
int_est <- round(
  coef(summary(model_interaction))["waterfront:view", "Estimate"],
  2
)
int_ci_lo <- round(interact_ci, 2)[1]
int_ci_hi <- round(interact_ci, 2)[2]

kable(
  pf_interact, digits = 4,
  caption = "Partial F-test: additive versus waterfront x view interaction model"
)

```

Partial F-test: additive versus waterfront x view interaction model

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
21519	1.003478e+15	NA	NA	NA	NA
21518	1.003428e+15	1	49854992641	1.0691	0.3012

Diagnostics

```

diag_data <- augment(final_model)
diag_data$sqrt_abs_std_resid <- sqrt(abs(diag_data$.std.resid))
diag_data$index <- seq_len(nrow(diag_data))

leverage_cutoff <- 2 * length(coef(final_model)) / nrow(diag_data)
cook_cutoff <- 4 / nrow(diag_data)
top_influence <- diag_data[
  order(diag_data$.cooks, decreasing = TRUE)[1:3],
]

```

```

plot_2x2 <- function(p1, p2, p3, p4) {
  grid::grid.newpage()
  grid::pushViewport(grid::viewport(layout = grid::grid.layout(2, 2)))
  print(p1, vp = grid::viewport(layout.pos.row = 1, layout.pos.col = 1))
  print(p2, vp = grid::viewport(layout.pos.row = 1, layout.pos.col = 2))
  print(p3, vp = grid::viewport(layout.pos.row = 2, layout.pos.col = 1))
  print(p4, vp = grid::viewport(layout.pos.row = 2, layout.pos.col = 2))
}

p_resid_fit <- ggplot(diag_data, aes(x = .fitted, y = .resid)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey45") +
  geom_point(size = 1.7, alpha = 0.8, color = "#1B4F72") +
  geom_smooth(
    method = "loess", formula = y ~ x,
    se = FALSE, color = "#D55E00", linewidth = 0.8
  ) +
  labs(title = "Residuals vs Fitted", x = "Fitted values", y = "Residuals")

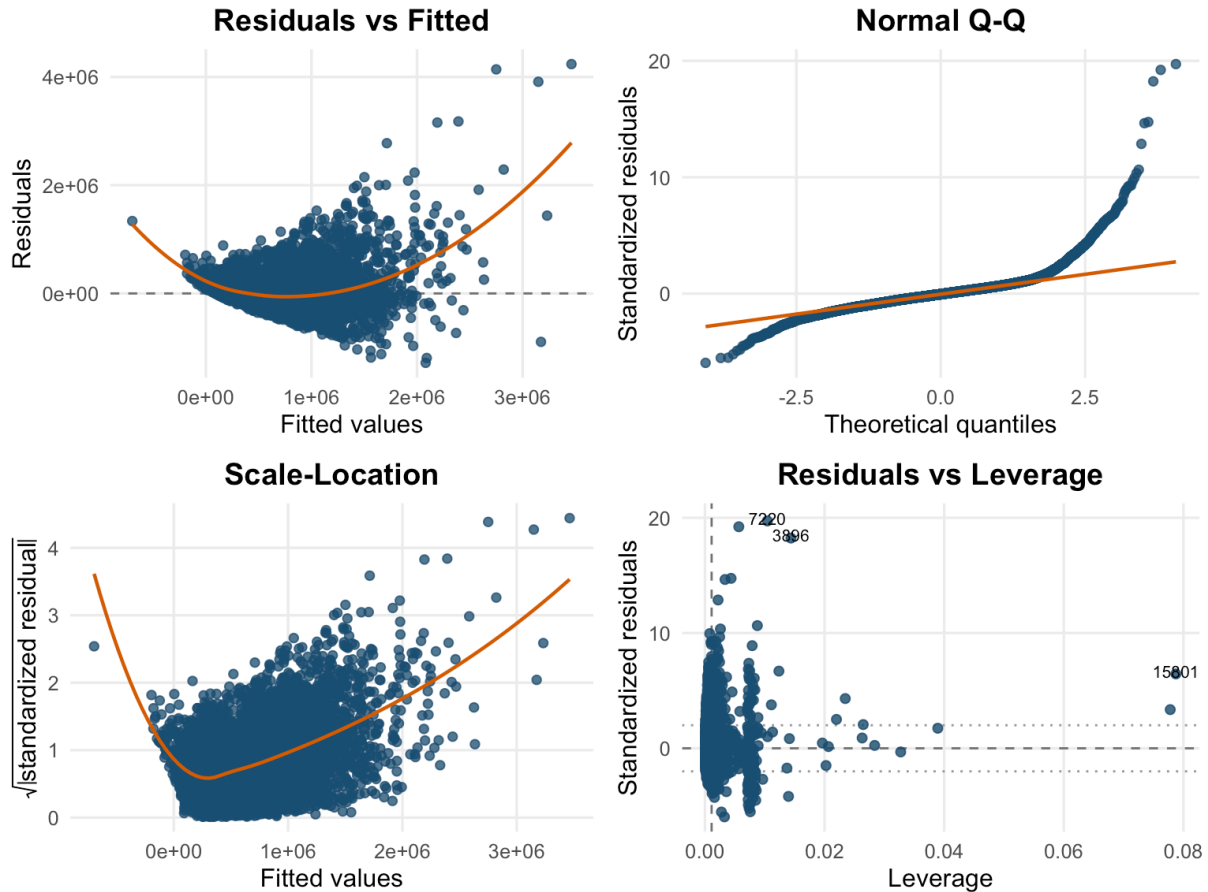
p_qq <- ggplot(diag_data, aes(sample = .std.resid)) +
  stat_qq(size = 1.7, alpha = 0.8, color = "#1B4F72") +
  stat_qq_line(color = "#D55E00", linewidth = 0.8) +
  labs(
    title = "Normal Q-Q",
    x = "Theoretical quantiles", y = "Standardized residuals"
  )

p_scale <- ggplot(diag_data, aes(x = .fitted, y = sqrt_abs_std_resid)) +
  geom_point(size = 1.7, alpha = 0.8, color = "#1B4F72") +
  geom_smooth(
    method = "loess", formula = y ~ x,
    se = FALSE, color = "#D55E00", linewidth = 0.8
  ) +
  labs(
    title = "Scale-Location", x = "Fitted values",
    y = expression(sqrt(abs("standardized residual")))
  )

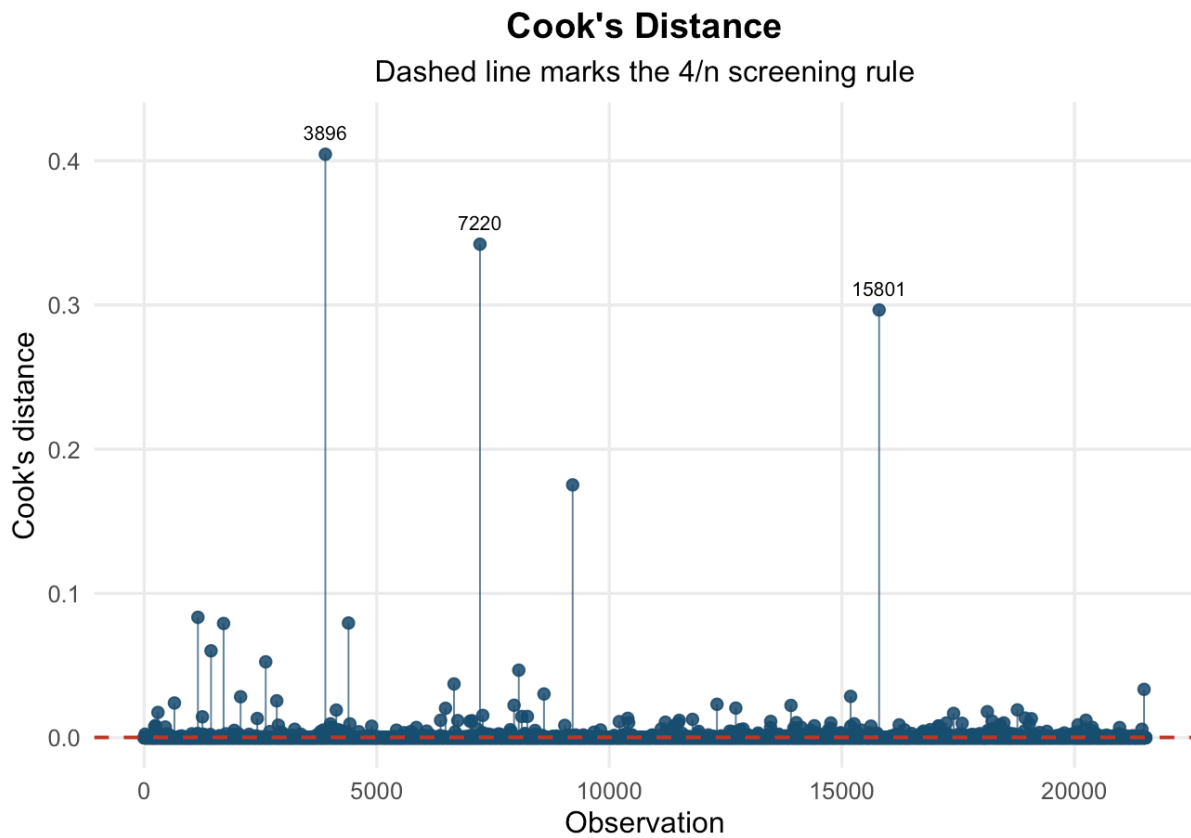
p_leverage <- ggplot(diag_data, aes(x = .hat, y = .std.resid)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey45") +
  geom_hline(yintercept = c(-2, 2), linetype = "dotted", color = "grey60") +
  geom_vline(xintercept = leverage_cutoff, linetype = "dashed", color = "grey45") +
  geom_point(size = 1.8, alpha = 0.85, color = "#1B4F72") +
  geom_text(
    data = top_influence, aes(label = index),
    nudge_y = 0.25, size = 2.8, show.legend = FALSE
  ) +
  labs(
    title = "Residuals vs Leverage",
    x = "Leverage", y = "Standardized residuals"
  )

```

```
plot_2x2(p_resid_fit, p_qq, p_scale, p_leverage)
```



```
ggplot(diag_data, aes(x = index, y = .cooksd)) +
  geom_segment(
    aes(xend = index, y = 0, yend = .cooksd),
    linewidth = 0.35, alpha = 0.75, color = "#1B4F72"
  ) +
  geom_point(size = 1.9, alpha = 0.9, color = "#1B4F72") +
  geom_hline(
    yintercept = cook_cutoff, linetype = "dashed",
    color = "#C0392B", linewidth = 0.7
  ) +
  geom_text(
    data = top_influence, aes(label = index),
    nudge_y = 0.015, size = 2.8, show.legend = FALSE
  ) +
  labs(
    title = "Cook's Distance",
    subtitle = "Dashed line marks the 4/n screening rule",
    x = "Observation",
    y = "Cook's distance"
  )
```



```
set.seed(123)

shapiro_result <- shapiro.test(sample(residuals(final_model), 5000))
bp_result <- lmtest::bptest(final_model)
bp_stat <- as.numeric(bp_result$statistic)
bp_p <- bp_result$p.value

dw_result <- lmtest::dwtest(final_model)
dw_stat <- as.numeric(dw_result$statistic)
dw_p <- dw_result$p.value

diagnostic_table <- data.frame(
  Assumption = c("Normality", "Constant Variance", "Independence"),
  Test = c("Shapiro-Wilk", "Breusch-Pagan", "Durbin-Watson"),
  Statistic = round(c(
    as.numeric(shapiro_result$statistic),
    bp_stat,
    dw_stat
  ), 4),
  `p-value` = c(
    shapiro_result$p.value,
    bp_p,
    dw_p
  ),
  check.names = FALSE
)

kable(diagnostic_table, caption = "Formal Assumption Tests")
```

Formal Assumption Tests

Assumption	Test	Statistic	p-value
Normality	Shapiro-Wilk	0.8307	0.0000000
Constant Variance	Breusch-Pagan	2603.4436	0.0000000
Independence	Durbin-Watson	1.9774	0.0486924